

A Source-Controlled Nonlinear Kapitza Model

Exact reduction, imperfect pitchfork unfolding, and S-curves

Detailed report accompanying the Python and MATLAB implementations

Abstract

This report develops a one-dimensional nonlinear Kapitza transmission model in which the normalized strength κ of a positive volume heat source is the bifurcation parameter. The underlying two-layer Poisson problem is the one-dimensional non-uniqueness example in the recent work of Bejaoui, Ben Belgacem, Jelassi, and Ma. Eliminating the two piecewise temperature fields gives the jump relation $s = [u] = \kappa - (1 + \delta)a$, where $\delta = \lambda/\mu$ is the ratio of the bulk conductivities. A specifically constructed exponential contact conductance makes the scalar equilibrium equation exactly equivalent to $x = \tanh(\kappa x + h)$. Consequently, the symmetric problem undergoes a supercritical pitchfork at the critical source strength $\kappa_c = 1$. Positive and negative biases unfold the pitchfork, the saddle-node trajectory is analytic, and its local displacement satisfies a two-thirds power law. The report also reconstructs the physical temperature fields, distinguishes Picard-iteration stability from physical time-dependent stability, and documents matching single-file Python and MATLAB implementations.

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1 The source problem from the two-layer Poisson model

Consider the composite interval

$$\Omega = [-1, 1], \quad \Omega_I = [-1, 0], \quad \Omega_E = [0, 1], \quad (1)$$

with a single interface at $z = 0$. Let $\lambda > 0$ and $\mu > 0$ be the constant thermal conductivities in Ω_I and Ω_E , respectively. The endpoint temperatures are homogeneous,

$$u_I(-1) = 0, \quad u_E(1) = 0, \quad (2)$$

and the positive volume source is supported only in the right layer:

$$f(z) = 2\mu\kappa \chi_{\Omega_E}(z), \quad \kappa > 0. \quad (3)$$

The steady bulk equations are therefore

$$-\lambda u_I'' = 0 \quad \text{in } \Omega_I, \quad -\mu u_E'' = 2\mu\kappa \quad \text{in } \Omega_E. \quad (4)$$

This resolves an important notation issue. In this model κ is not a bulk conductivity: the conductivities are λ and μ . After the right equation in (4) is divided by μ , the normalized source amplitude is 2κ . Thus a blackboard notation such as $\alpha = 2\kappa$ should be read as a source-amplitude relation, not as the old numerical sigmoid-slope convention. The general paper also uses the letter α for the piecewise conductivity field, so the symbols must be kept separate.

2 Exact elimination of the bulk temperature fields

Solving (4) and imposing the exterior boundary conditions gives

$$u(z) = u_{(a,b)}(z) = \begin{cases} a(z+1), & z \in \Omega_I, \\ b(z-1) + \kappa(1-z^2), & z \in \Omega_E. \end{cases} \quad (5)$$

The interface traces are

$$u_I(0) = a, \quad u_E(0) = \kappa - b. \quad (6)$$

Flux conservation and the nonlinear Kapitza law take the form

$$\lambda a = \mu b = \beta_{\text{phys}}(a, \kappa - b)(\kappa - b - a). \quad (7)$$

Introduce the conductivity ratio

$$\delta := \frac{\lambda}{\mu} > 0. \quad (8)$$

Flux continuity gives $b = \delta a$. With the jump convention $s = [u] = u_E(0) - u_I(0)$, one obtains the exact bulk relation

$$\boxed{s = \kappa - (1 + \delta)a.} \quad (9)$$

Define the conductance and flux normalized by μ ,

$$\hat{\beta} := \frac{\beta_{\text{phys}}}{\mu}, \quad \hat{q} := \frac{q}{\mu}. \quad (10)$$

Then (7) becomes

$$\boxed{\hat{q} = \delta a = \hat{\beta}(s)s.} \quad (11)$$

Solving (9)–(11) for a produces the scalar map

$$a = F_{\kappa,h}(a) := \frac{\kappa \hat{\beta}_{\kappa,h}(s)}{\delta + (1 + \delta) \hat{\beta}_{\kappa,h}(s)}, \quad s = \kappa - (1 + \delta)a. \quad (12)$$

Because $\hat{\beta} > 0$, a positive steady heat flux gives $a > 0$ and $s > 0$. Hence the physical interval moves with the source parameter:

$$0 < a < \gamma(\kappa), \quad \gamma(\kappa) := \frac{\kappa}{1 + \delta}, \quad 0 < s < \kappa. \quad (13)$$

The source term is essential. If both layers were source-free while the two exterior temperatures were zero and the conductances were positive, the standard energy identity would force the trivial solution.

3 Exponential Kapitza law and normalized fixed-point equation

The new constitutive choice is

$$\boxed{\hat{\beta}_{\kappa,h}(s) = \frac{\delta}{1 + \delta} \exp[2(\kappa - 2s) + 2h].} \quad (14)$$

It is positive, decreasing, and convex as a function of the jump:

$$\partial_s \hat{\beta}_{\kappa,h} = -4 \hat{\beta}_{\kappa,h} < 0, \quad \partial_{ss} \hat{\beta}_{\kappa,h} = 16 \hat{\beta}_{\kappa,h} > 0. \quad (15)$$

The bias has a direct constitutive interpretation:

$$\hat{\beta}_{\kappa,h}(s) = e^{2h} \hat{\beta}_{\kappa,0}(s). \quad (16)$$

Define the normalized state

$$x := \frac{2a}{\gamma(\kappa)} - 1 = \frac{2(1 + \delta)a}{\kappa} - 1 = 1 - \frac{2s}{\kappa}, \quad -1 < x < 1. \quad (17)$$

Writing $E = \exp[2(\kappa - 2s) + 2h]$, equation (12) becomes

$$F_{\kappa,h}(a) = \frac{\gamma(\kappa)}{2} [1 + \tanh(\kappa - 2s + h)]. \quad (18)$$

Since $\kappa - 2s = \kappa x$, the complete transmission problem is exactly equivalent to

$$x = \tanh(\kappa x + h). \quad (19)$$

The material ratio δ remains in the conversion back to a , but it drops out of the normalized root equation.

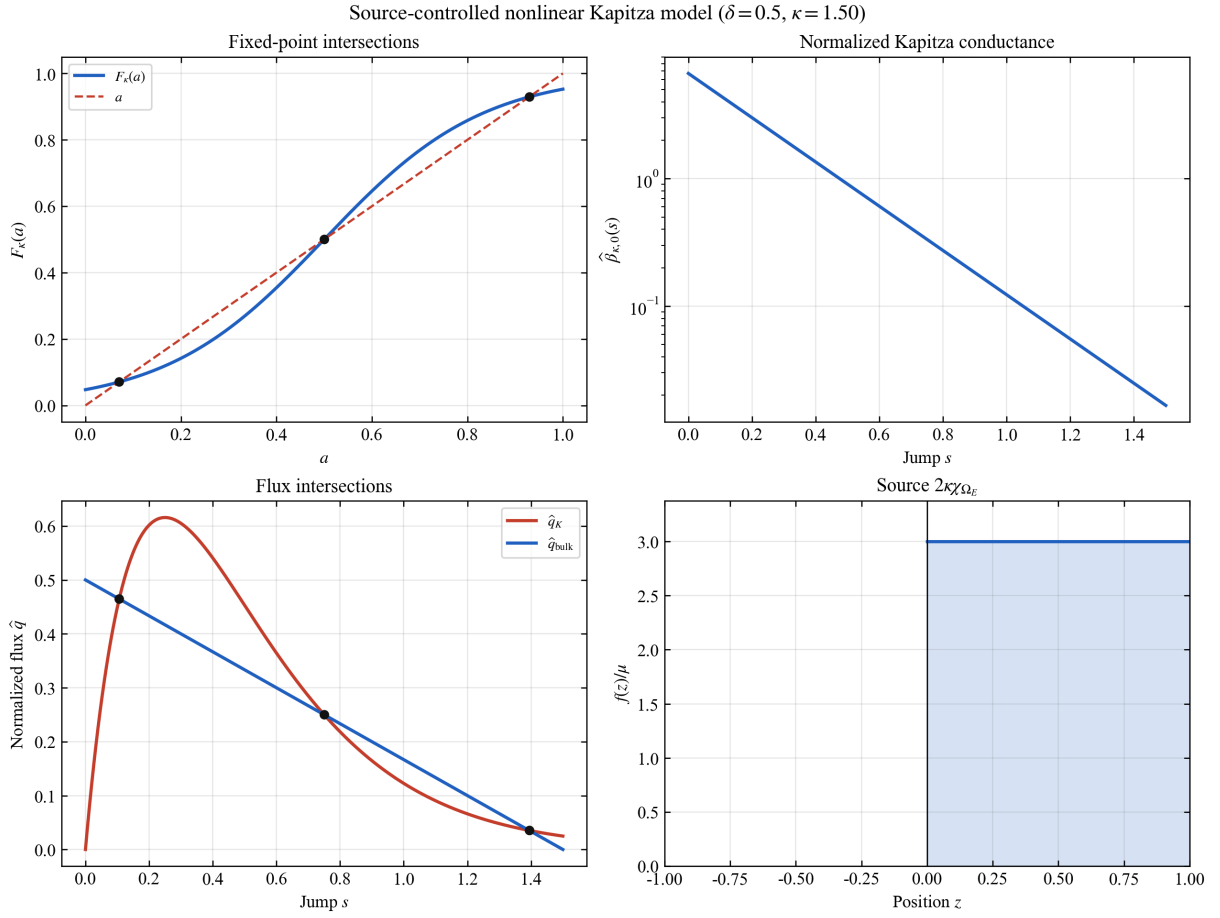


Figure 1: The source, nonlinear conductance, scalar map, and heat-flux intersections are four views of the same reduced transmission problem.

4 Symmetric source-controlled pitchfork

Set $h = 0$ and define $G(x) = \tanh(\kappa x) - x$. The central root $x = 0$ is always present and

$$G'(0) = \kappa - 1. \quad (20)$$

It follows that

$$\begin{array}{ll} 0 < \kappa \leq 1 : & \text{one steady state,} \\ \kappa > 1 : & \text{three steady states.} \end{array} \quad (21)$$

Thus the critical source strength is

$$\kappa_c = 1. \quad (22)$$

The central branch and its physical variables are

$$x_c = 0, \quad a_c(\kappa) = \frac{\kappa}{2(1+\delta)}, \quad s_c(\kappa) = \frac{\kappa}{2}. \quad (23)$$

Consequently, the central branch is horizontal in (κ, x) , but it is a linearly increasing curve in the physical (κ, a) plane. For $x \neq 0$, the two outer branches have the exact parameterization

$$\kappa(x) = \frac{\operatorname{artanh} x}{x}, \quad a(x) = \frac{\kappa(x)(1+x)}{2(1+\delta)}, \quad 0 < |x| < 1. \quad (24)$$

Since

$$\frac{\operatorname{artanh} x}{x} = 1 + \frac{x^2}{3} + O(x^4), \quad (25)$$

the outer states satisfy $x \sim \pm\sqrt{3(\kappa - 1)}$: this is a supercritical pitchfork.

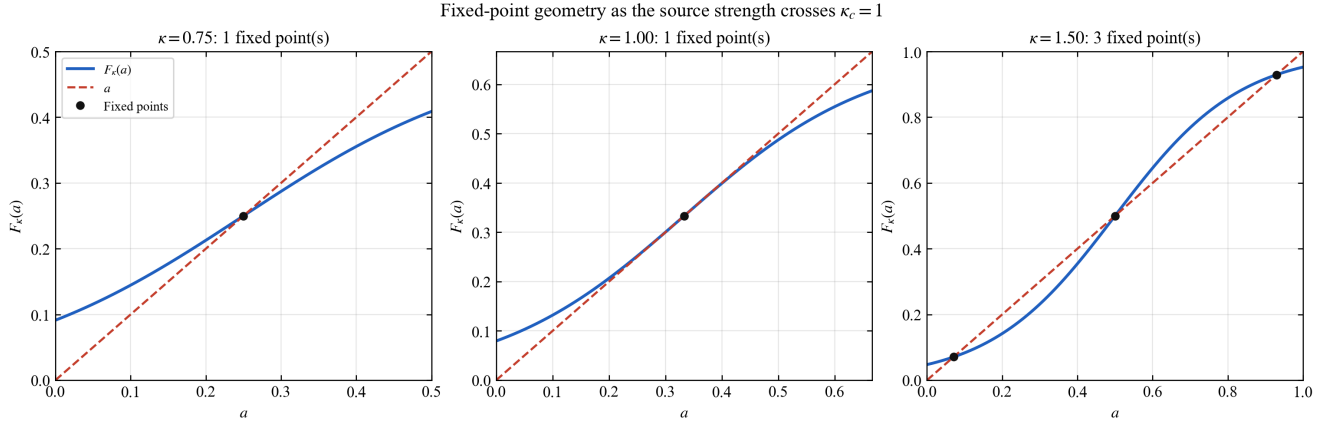


Figure 2: Fixed-point intersections below, at, and above the critical source strength. Each panel uses its own physical interval $0 < a < \kappa/(1 + \delta)$.

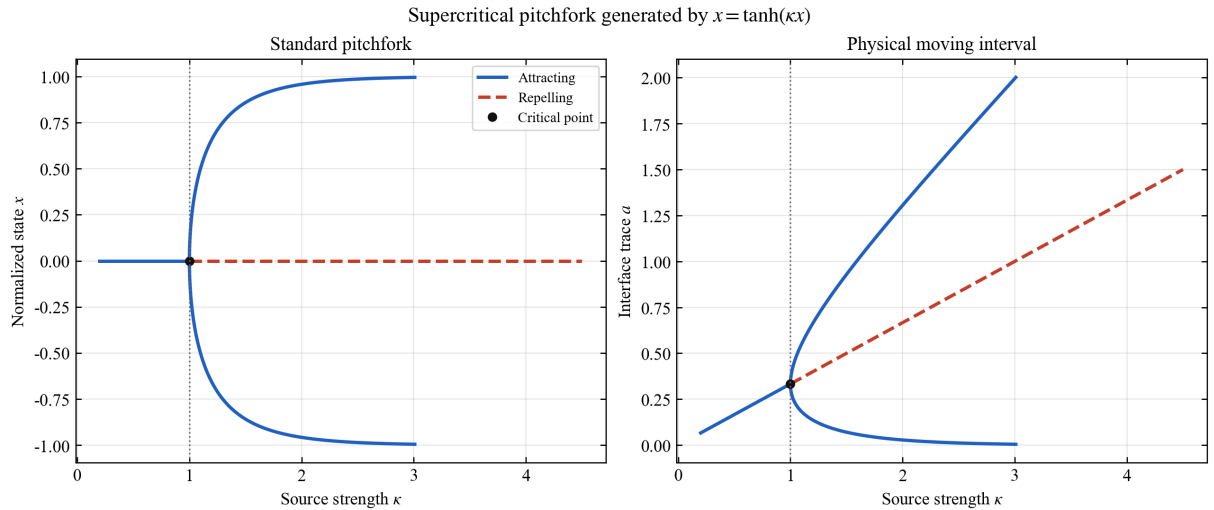


Figure 3: The standard pitchfork in normalized coordinates and the same branches reconstructed in the physical (κ, a) plane.

5 Picard stability and cobweb interpretation

The numerical fixed-point iteration is

$$x_{n+1} = \tanh(\kappa x_n + h). \quad (26)$$

At a fixed point, its multiplier is

$$m_{\text{FP}} = \kappa \operatorname{sech}^2(\kappa x + h) = \kappa(1 - x^2). \quad (27)$$

The notation m_{FP} avoids confusion with the left conductivity λ . A root is locally attracting for this scalar iteration when $|m_{\text{FP}}| < 1$, and repelling when $|m_{\text{FP}}| > 1$.

For $h = 0$, the central multiplier is $m_{\text{FP}} = \kappa$. Therefore the central branch is attracting for $\kappa < 1$ and repelling for $\kappa > 1$. The two outer branches are attracting. At $\kappa = 1$, the linearization is neutral, but $\tanh x = x - x^3/3 + O(x^5)$, so nearby iterations approach the origin only algebraically. This produces the visible critical slowing in the cobweb plot.

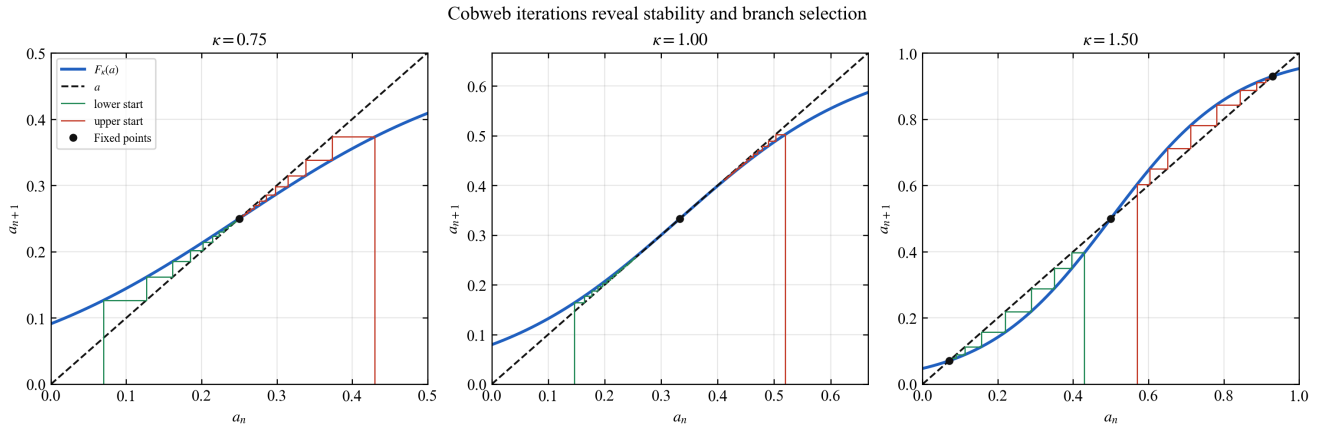


Figure 4: Cobweb paths reveal convergence to the center below threshold, critical slowing at $\kappa = 1$, and selection of the two attracting outer branches above threshold.

This classification concerns the Picard algorithm, equivalently the discrete-time map used to compute the fixed points. It is not automatically a proof of dynamical stability for a time-dependent heat equation; that would require specifying and linearizing a transient transmission model.

6 Nonzero bias and imperfect pitchfork unfolding

For a fixed nonzero bias, solving (19) for the control parameter gives the exact branch parameterization

$$\kappa(x; h) = \frac{\operatorname{artanh} x - h}{x}, \quad a(x; h) = \frac{\kappa(x; h)(1 + x)}{2(1 + \delta)}. \quad (28)$$

The symmetry is

$$(x, h) \mapsto (-x, -h), \quad (29)$$

so negative-bias diagrams are reflected versions of positive-bias diagrams in normalized state space. In physical coordinates the moving factor $\gamma(\kappa)$ must still be applied.

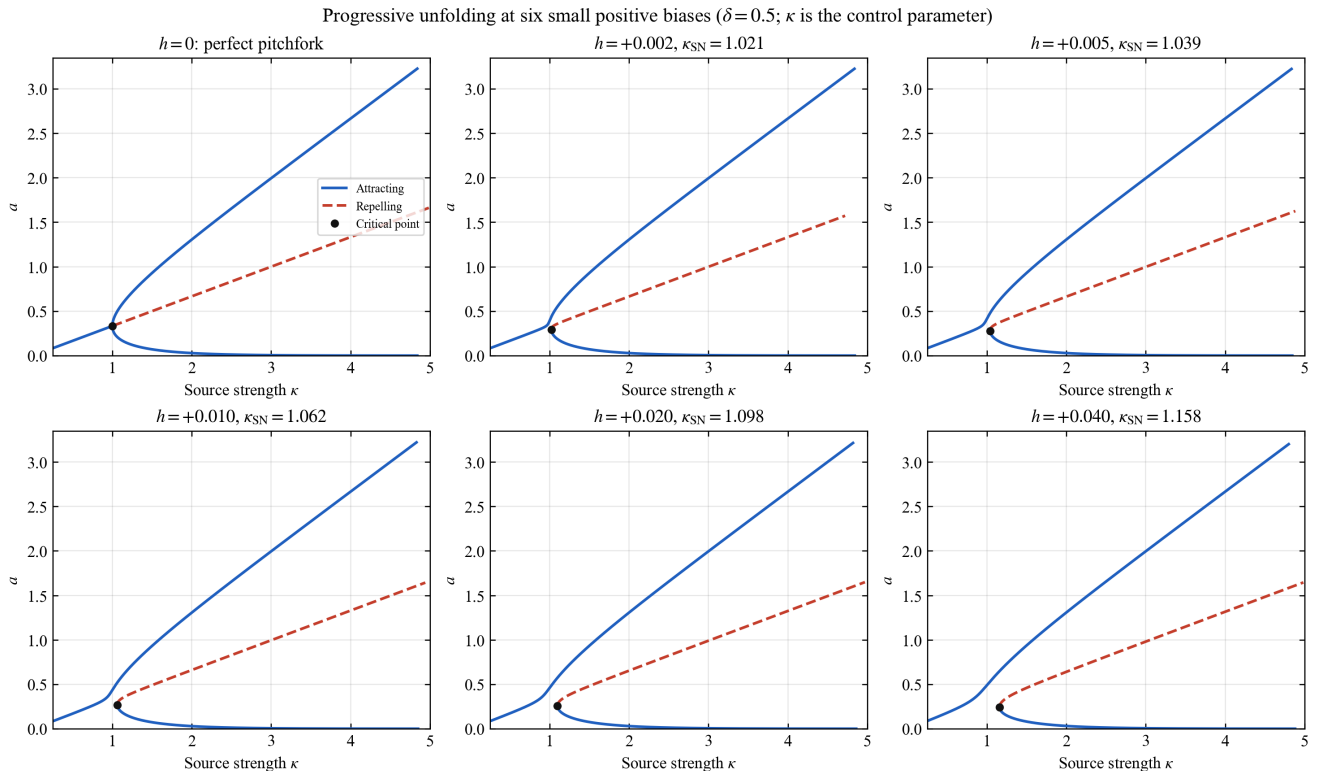


Figure 5: Six fixed positive biases with κ retained as the source control parameter. Solid branches are attracting for Picard iteration and dashed branches are repelling.

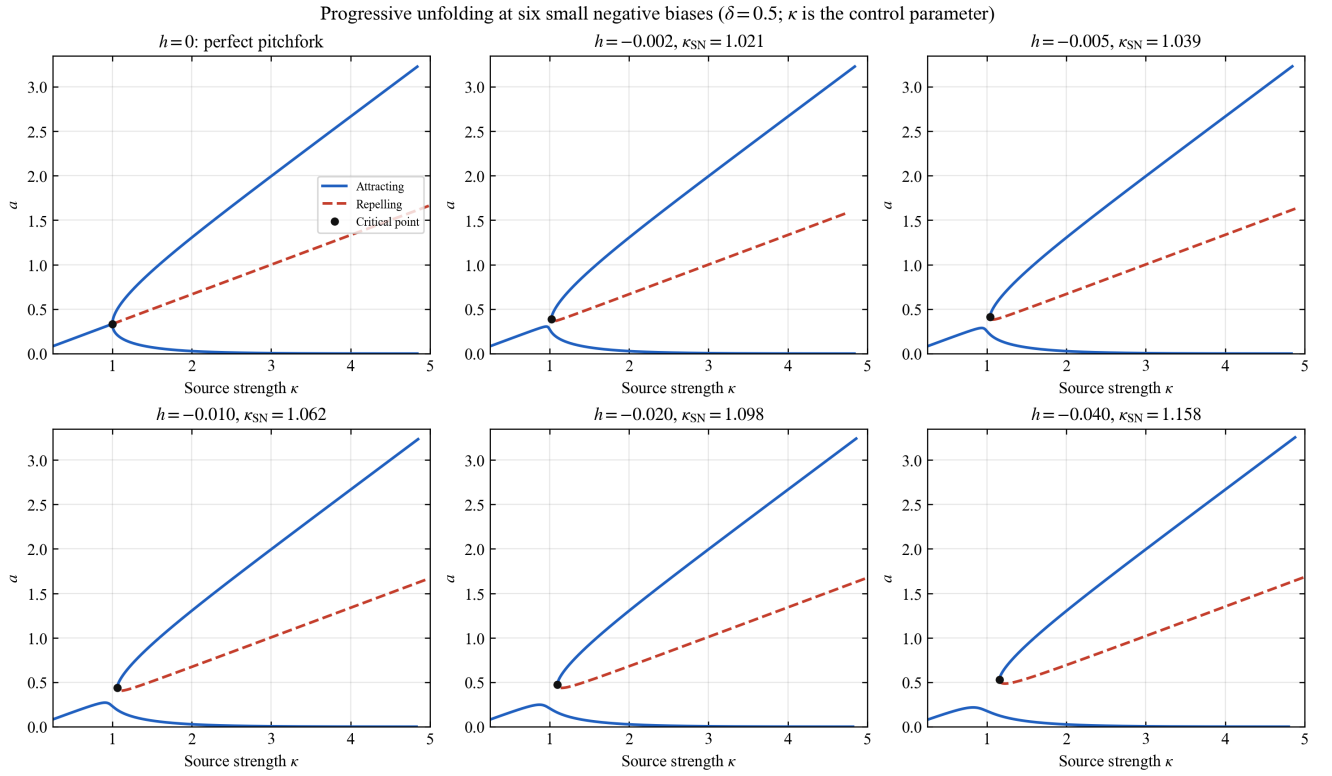


Figure 6: The corresponding six nonpositive biases. The preferred branch and the finite saddle node are reflected relative to the positive-bias case.

7 Saddle-node trajectory and local two-thirds law

At a fold, the residual and its derivative vanish. Combining

$$x = \tanh(\kappa x + h), \quad 1 = \kappa(1 - x^2), \quad (30)$$

gives

$$\kappa_{\text{SN}} = \frac{1}{1 - x_{\text{SN}}^2}, \quad h = \text{artanh}(x_{\text{SN}}) - \frac{x_{\text{SN}}}{1 - x_{\text{SN}}^2}. \quad (31)$$

The physical critical state is

$$a_{\text{SN}} = \frac{\kappa_{\text{SN}}(1 + x_{\text{SN}})}{2(1 + \delta)}. \quad (32)$$

Near the perfect pitchfork,

$$h = -\frac{2}{3}x_{\text{SN}}^3 + O(x_{\text{SN}}^5), \quad (33)$$

$$\kappa_{\text{SN}} - 1 = x_{\text{SN}}^2 + O(x_{\text{SN}}^4). \quad (34)$$

Eliminating x_{SN} gives the local cusp scaling

$$\kappa_{\text{SN}} - 1 \sim \left(\frac{3|h|}{2} \right)^{2/3}. \quad (35)$$

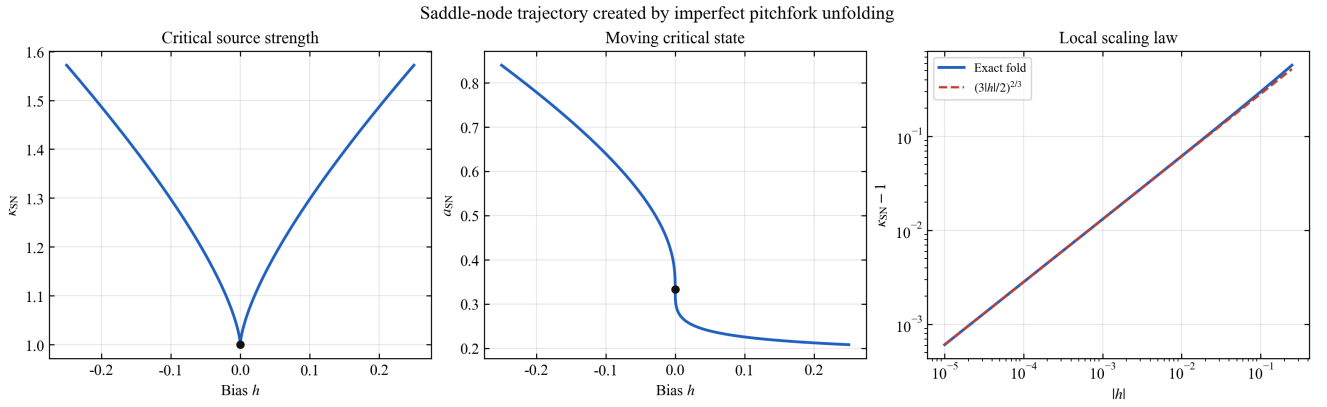


Figure 7: The critical source strength, the physical fold state, and the agreement between the exact saddle-node trajectory and the local two-thirds law.

8 Fixed-source S-curve and heat-flux intersections

For a fixed $\kappa > 1$, use h as the continuation parameter. The S-curve has the exact physical parameterization

$$h(x) = \operatorname{artanh} x - \kappa x, \quad a(x; \kappa) = \frac{\kappa(1+x)}{2(1+\delta)}. \quad (36)$$

The two fold states are

$$x_f = \pm \sqrt{1 - \frac{1}{\kappa}}, \quad (37)$$

with bias magnitude

$$h_c = \kappa \sqrt{1 - \frac{1}{\kappa}} - \operatorname{artanh} \sqrt{1 - \frac{1}{\kappa}}. \quad (38)$$

The same multiplicity can be displayed as heat-flux intersections. Eliminating a from (9) gives the bulk-required normalized flux

$$\hat{q}_{\text{bulk}}(s; \kappa) = \frac{\delta}{1+\delta}(\kappa - s), \quad (39)$$

whereas the interface law supplies

$$\hat{q}_K(s; \kappa, h) = \hat{\beta}_{\kappa, h}(s)s. \quad (40)$$

Their intersections are exactly the steady states.

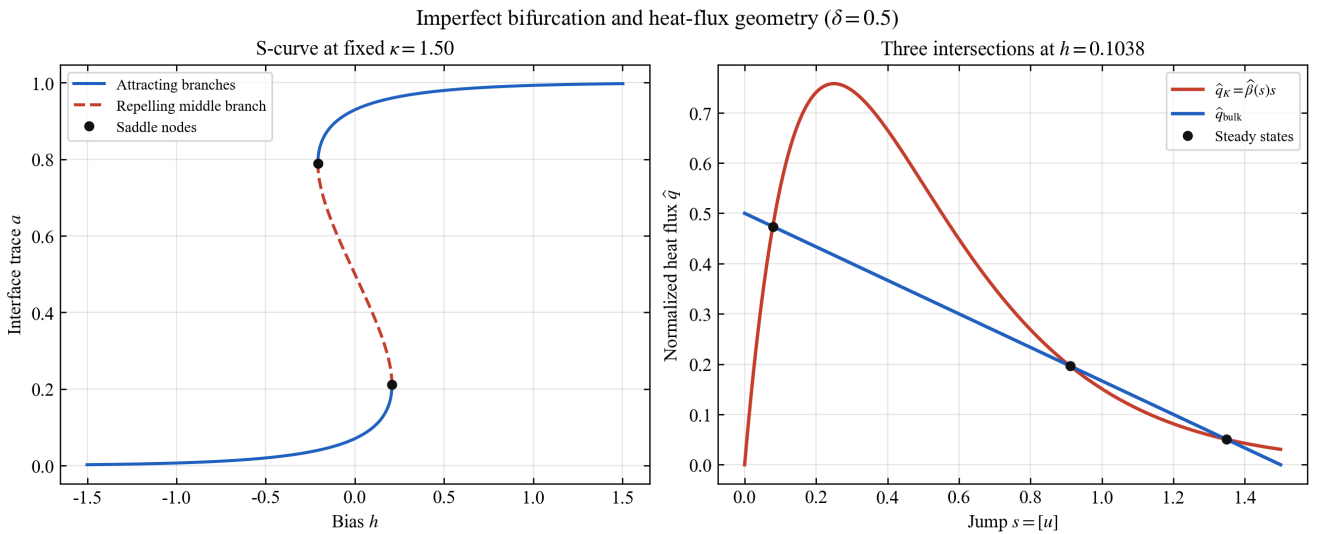


Figure 8: For $\delta = 0.5$ and $\kappa = 1.5$, the fixed-source S-curve has two analytic saddle nodes. An interior fixed-bias slice has three equivalent heat-flux intersections.

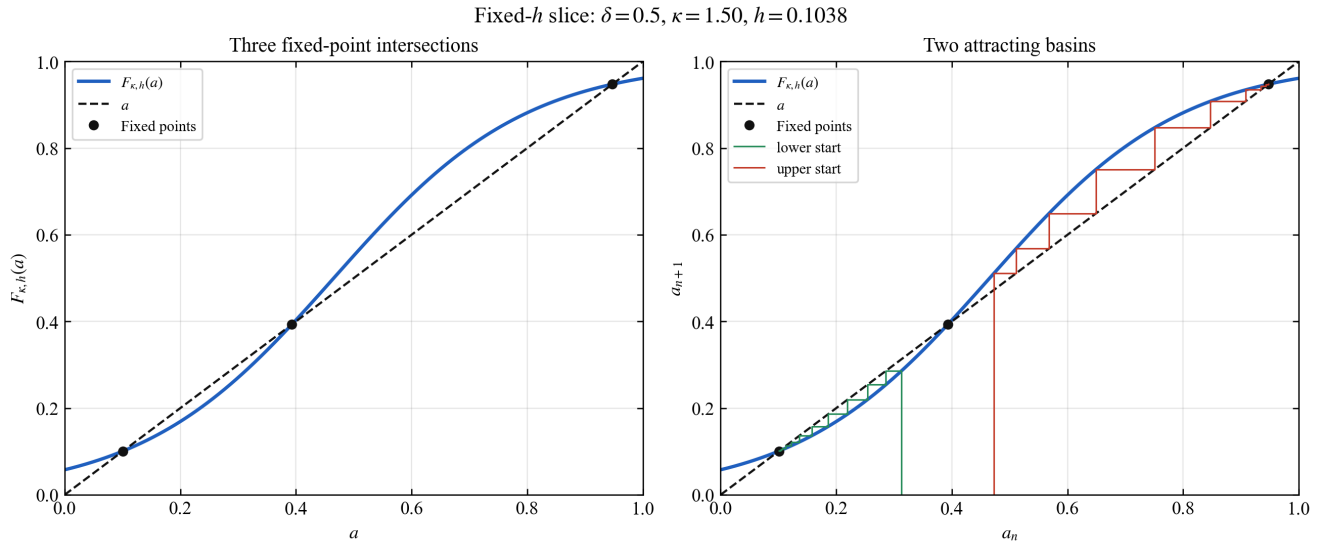


Figure 9: At a fixed interior bias, the three fixed-point intersections and two initial states on opposite sides of the repelling middle root display the two attracting basins.

9 Reconstruction of the piecewise temperature fields

Each scalar root gives a complete steady transmission solution through (5). For the representative symmetric case $\delta = 0.5, \kappa = 1.5$, the normalized roots are approximately

$$x = -0.8585596366, \quad x = 0, \quad x = 0.8585596366. \quad (41)$$

The corresponding left interface traces are

$$a \approx 0.0707201817, \quad 0.5, \quad 0.9292798183. \quad (42)$$

The discontinuities at $z = 0$ are not numerical artifacts: they are the temperature jumps permitted by the nonlinear Kapitza contact resistance.

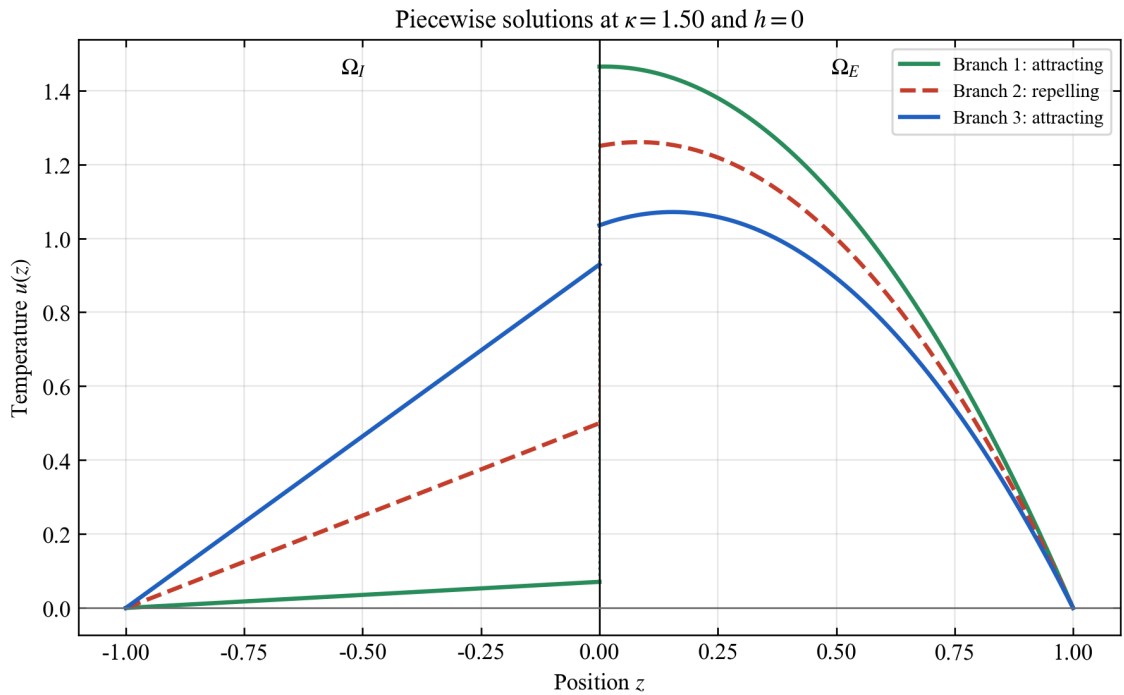


Figure 10: The three complete piecewise solutions reconstructed from the three scalar fixed points. Solid and dashed styles retain the Picard stability convention.

10 Numerical implementation and reproducibility

The Python and MATLAB programs implement the same workflow:

1. solve $x = \tanh(\kappa x + h)$ by exact symmetry when $h = 0$ and by robust bracketed root isolation when $h \neq 0$;
2. reconstruct $a, b, s, \hat{\beta}, \hat{q}$ and the multiplier from each root;
3. parameterize the perfect and imperfect branches analytically;
4. encode attracting segments as solid curves and repelling segments as dashed curves;
5. generate matching PNG and vector PDF figures, plus CSV state tables.

The Python entry point is

```
python3 code/python/kapitza_kappa_model.py
```

when executed from the project directory. Optional arguments are `-delta`, `-kappa`, and `-output`.

The MATLAB entry point is

```
run('code/matlab/kapitza_kappa_model.m')
```

and requires no toolbox beyond standard MATLAB plotting and root finding. The two programs are deliberately single-file implementations with English comments throughout. The MATLAB script also accepts the environment variable `KAPITZA_STAGE=early` or `late` for systems that impose a short batch-execution limit; without it, the complete workflow runs normally.

11 Interpretation and scope

The completed analysis separates three logically distinct components:

- The two-layer Poisson equation, its volume source, the piecewise solution, and the scalar interface reduction come from the cited one-dimensional transmission example.
- The exponential conductance (14) is a new constitutive construction placed on that confirmed reduction. The source paper's principal bifurcation examples instead use power-law conductances.
- Stability in the present figures means local attraction or repulsion of the Picard fixed-point iteration. A claim about physical transient stability would require a separate time-dependent model.

Within this scope, κ is now a genuine physical control parameter: it is the normalized amplitude of the positive source in Ω_E . The value $\kappa_c = 1$ is a critical normalized source strength, not a condition that the two material conductivities are equal. Material contrast is measured by $\delta = \lambda/\mu$.

Source

E. Bejaoui, F. Ben Belgacem, F. Jelassi, and J. Ma, *Semilinear Elliptic Problems with NonLinear Contact Resistance in Composite Domains*, HAL preprint, Section 7, [hal-05659943v1](https://hal.archives-ouvertes.fr/hal-05659943v1).